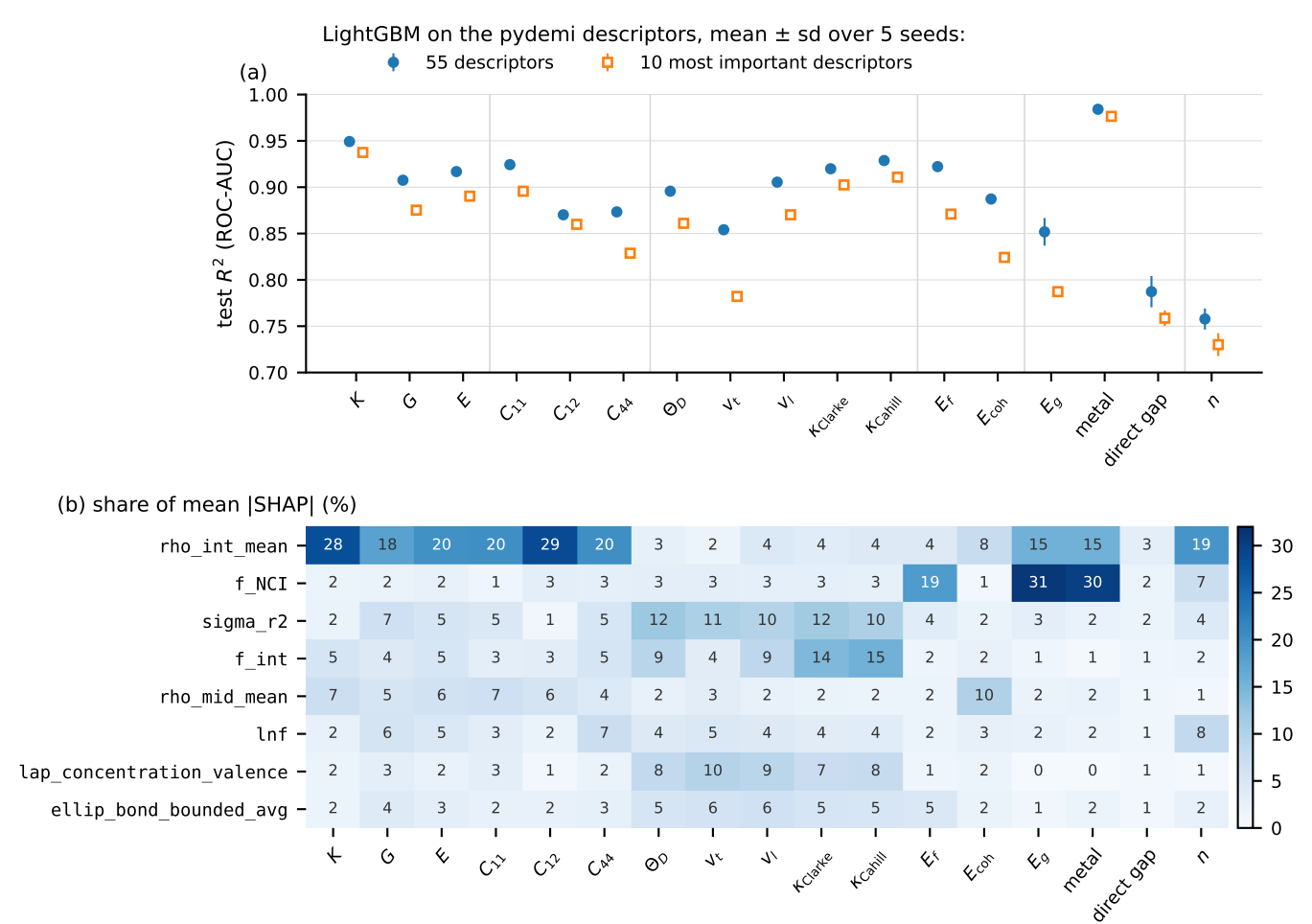
LightGBM on the pydemi descriptors, mean ± sd over 5 seeds:
(a)
55 descriptors
10 most important descriptors
test R^2 (ROC-AUC)
1.00
0.95
0.90
0.85
0.80
0.75
0.70
K
G
E
C_{11}
C_{12}
C_{44}
Θ_D
v_t
v_l
κ_{Clarke}
κ_{Cahill}
E_f
E_{coh}
E_g
metal
direct gap
n
(b) share of mean |SHAP| (%)
rho_int_mean 28 18 20 20 29 20 3 2 4 4 4 4 8 15 15 3 19
f_NCI 2 2 2 1 3 3 3 3 3 3 3 19 1 31 30 2 7
sigma_r2 2 7 5 5 1 5 12 11 10 12 10 4 2 3 2 2 4
f_int 5 4 5 3 3 5 9 4 9 14 15 2 2 1 1 1 2
rho_mid_mean 7 5 6 7 6 4 2 3 2 2 2 2 10 2 2 1 1
lnf 2 6 5 3 2 7 4 5 4 4 4 2 3 2 2 1 8
lap_concentration_valence 2 3 2 3 1 2 8 10 9 7 8 1 2 0 0 1 1
ellip_bond_bounded_avg 2 4 3 2 2 3 5 6 6 5 5 5 2 1 2 1 2
K
G
E
C_{11}
C_{12}
C_{44}
Θ_D
v_t
v_l
κ_{Clarke}
κ_{Cahill}
E_f
E_{coh}
E_g
metal
direct gap
n
30
25
20
15
10
5
0